MIL-C-10351D(GL)
26 November 1982
SUPERSEDING
MIL-C-10351C(GL)
24 June 1969

MILITARY SPECIFICATION

CLOTH, LAMINATED AND COATED FOR WATERPROOF CONTAINERS

This specification is approved for use by the Natick Research and Development Laboratories, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers two types of two-ply, laminated rubber coated nylon cloth.

1.2 Classification. The laminated and coated cloth shall be of the following types as specified (see 6.2).

Type I - Throat cloth, two ply.

Type II - Body cloth, two-ply.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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SPECIFICATIONS

FEDERAL

PPP-P-1136 - Packaging and Packing of Coated (Plastic; Rubber) and Laminated Fabrics.

MILITARY

MIL-C-577 - Cloth, Twill, Nylon, 1.6 and 3.0 Ounce.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.
FED-STD-406 - Plastics: Methods of Testing.
FED-STD-601 - Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1487 - Glossary of Cloth Coating Imperfections.

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

* 3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.3).

3.2 Materials.

3.2.1 Base cloths.

3.2.1.1 Type I. The base cloth used to construct the type I laminated, coated cloth shall be nylon twill, any color, conforming to type I of MIL-C-577.

3.2.1.2 Type II. The base cloth used to construct the type II laminated, coated cloth shall be nylon twill, any color, conforming to type II of MIL-C-577.

3.2.2 Coating compound. The coating compound for both types shall be composed of natural or synthetic rubber or a mixture of both, suitably compounded and pigmented to meet the requirements herein. Reclaimed rubber shall not be used (see 4.4.1.1).

3.2.2.1 Color. Unless otherwise specified in the contract or purchase order, the color of the coating compound shall be black (see 6.2).

3.2.3 Laminated coated cloths. The laminated coated cloth shall be two-ply construction with rubber coating applied to both outside surfaces in approximately equal amounts and with a sufficient quantity applied between the plies to insure that the finished laminated cloth, when vulcanized, will meet the applicable requirements specified in table I.

* TABLE I. Physical requirements of laminated, coated cloths

Characteristics	Type I		Type II	
	Minimum	Maximum	Minimum	Maximum
Weight, ounces per square yard	8.0	12.5	10.5	14.5
Breaking strength, pounds:				
Initial:				
Warp	90	----	180	----
Filling	90	----	170	----
After accelerated weathering:				
Warp	70	----	160	----
Filling	70	----	150	----

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TABLE I. Physical requirements of laminated, coated cloths (cont'd)

Characteristics	Type I Minimum	Type I Maximum	Type II Minimum	Type II Maximum
Tearing strength, grams:				
Warp	576	----	1024	----
Filling	384	----	640	----
Hydrostatic resistance, p.s.i.:				
Initial	100	----	175	----
After accelerated weathering 1/	50	----	90	----
After resistance to low temperature 2/	80	----	140	----
Adhesion, pounds per 2-inch width:				
Surface coating	6.0	----	6.0	----
Ply	6.0	----	6.0	----
Stiffness, centimeters (warp only):				
At 70°F	12.0	15.0	13.0	17.0
At minus 65°F	----	20.0	-----	21.0
Abrasion resistance	3/	----	3/	----
Blocking, scale rating	----	(1)	-----	(1)

1/ The test specimens shall not become stiff and brittle or soft and tacky.

2/ There shall be no evidence of cracking or flaking of the coating on the specimens when tested as specified in 4.4.4.

3/ There shall be no visible loose fibers in the center 1-inch of the abraded portion of any specimen. Visible loose fibers shall be regarded as fibers from the nylon yarns used to weave the base cloth and not as entire yarns or yarn plies.

3.3 Length and put-up. The laminated and coated cloth shall be put up in rolls as specified in PPP-P-1136. The minimum length of any one roll shall be 40 continuous yards and the maximum number of pieces in any one roll shall be three. The laminated and coated cloth may be supplied in one continuous length containing splices providing that these splices are on one side of the laminated and coated cloth only and that any splice is not closer than 40 yards to any other splice.

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Splices shall be produced by a butted-type seam and shall be flagged so that their location in the roll can be readily seen. When a roll contains more than one piece, no piece shall be less than 25 yards in length and the ends of the pieces shall be overlapped and not joined by a seam.

3.4 Width. The minimum overall width shall be as specified (see 6.2). Both selvages shall be trimmed to give straight, uniform, fully laminated edges.

* 3.5 Odor. The coated cloth shall be free from any objectionable odor. Characteristic odor of coating is not considered objectionable.

3.6 Workmanship. The finished laminated and coated cloth shall conform to the quality of product established by this document, and the occurence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and tested for the characteristics specified in table III. The presence of any defect or failure of any test shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Certification. A certificate of compliance shall be furnished with each shipment or lot stating that the laminating and coating compound is natural or synthetic rubber or a combination of both and that no reclaimed rubber has been used.

* 4.4.2. End item yard-by-yard examination. The required yardage of each roll in the sample shall be examined on one side only; however, the side shall be alternated so that alternate rolls are examined on the face and back successively. The cloth shall be examined for the defects listed below. The same yardage shall be given a through-light inspection for pinholes and thinly coated areas. The through-lighting inspection shall be performed in accordance with MIL-STD-1487. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit shall be 1 linear yard. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects. The number of rolls from which the sample is to be selected shall be in accordance with table II. An approximately equal number of yards shall be examined from each roll in the sample.

* Examine	Defect	Classification Major	Minor
Coating and lamination	Not coated on both sides as specified	X	
	Any pinhole	X	
	Any uncoated area	X	
	Any cut, hole, tear, scratch, or abrasion mark that results in baring the base cloth.	X	
	Any surface scratch or abrasion mark that does not result in baring the base cloth.		X
	Uneven coating, thin area where coating is missing or noticeably thinner	X	
	Any lump or heavily coated area	X	
	Any pit, blister, tunnel, or delamination of coating	X	

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Examine	Defect	Classification Major	Minor
Coating and lamination (cont'd)	Any separation of the laminated cloths	X	
	Crease or wrinkle resulting in doubleing or adhesion of surfaces that cannot be corrected by manual pressure	X	
	Cracked or checked coating	X	
	Any tackiness (coating will adhere and not unroll readily)	X	
Color	Coating compound not black or not as specified	X	
	Uneven or mottled	X	
Width	Trimmed width less than specified		X
	Edges unlaminated	X	
Workmanship	Selvage not trimmed, resulting in non-uniform edge	X	
	Not clean		X
	Curled, folded, scalloped, or rolled edges	X	
	Any spot, stain, or streak more than 1 inch in combined directions. 1/	X	
	Baggy, wavy, or puckered, will not stay flat upon normal pressure.		X
	Objectionable odor. 2/		X

1/ Clearly visible at normal inspection distance (approximately 3-feet).

2/ Odors of chemicals commonly used in coating compounds shall not be regarded as objectionable.

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TABLE II. Examination criteria

Lot size (yards)	Sample size (rolls)	Maximum number of defects acceptable in sample 2/
Up to 1200 inclusive 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

2/ Applicable to length examination only (see 4.4.3.1).

4.4.3 Length examination.

* 4.4.3.1 Individual roll length examination. Each roll selected for yard-by-yard examination shall be examined for the defects listed below. If the total number of defects in the sample rolls exceeds the maximum number specified in table II, the lot shall be rejected.

Any roll containing more than three pieces.
Any roll with a total length of less than 40 yards.
Any piece in roll less than 25 yards.
Ends of pieces in roll not overlapped.
Ends of pieces in roll joined by a seam.
Any roll with splices on both sides (faces).
Any splice less than 40 yards from any other splice.
Splice not butted type seam, or not flagged so that its location in roll can be readily seen.
Any roll with a gross length more than 2 yards less than that marked on ticket.

4.4.3.2 Total yardage in sample. The rolls examined shall be those selected for examination of individual rolls as specified in 4.4.3.1. The lot shall be unacceptable if the total of the actual gross lengths of the rolls in the sample is less than the total gross lengths marked on the tickets.

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* 4.4.4 End item testing. The laminated and coated cloth shall be tested for the characteristics listed in table III. The methods of testing specified in FED-STD-191, whenever applicable and as listed in table III shall be followed. The physical values specified in section 3 apply to the average of determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3 continuous yards full width of the laminated and coated cloth. The lot size shall be expressed in units of 1 linear yard and the sample size (number of units) shall be as specified below. The lot shall be unacceptable if any unit fails to meet any requirement specified. All test reports shall contain the individual values used in expressing the final results.

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

4.4.4.1 Standard test conditions. Results of physical tests obtained under testing conditions defined in FED-STD-191, FED-STD-406, or FED-STD-601 will be acceptable except in cases of dispute. In dispute cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in FED-STD-191.

* TABLE III. End item tests

Characteristic	Requirement paragraph	Test method
Weight	3.2.3	5041
Breaking strength:		
Initial	3.2.3	5102
After accelerated weathering	3.2.3	5804 1/ and 5102
Tearing strength	3.2.3	ASTM D 1424 2/
Hydrostatic resistance:		
Initial	3.2.3	5512 3/
After accelerated weathering	3.2.3	5804 1/ and 5512 4/
After resistance to low temperature	3.2.3	5874 and 5512 5/

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* TABLE III. End item tests (cont'd)

Characteristic	Requirement paragraph	Test method
Adhesion		
Of surface coating (either side)	3.2.3	5970
Of ply	3.2.3	5950
Stiffness (warp only)		
At 70° ± 2°F	3.2.3	5204
At - 65° ± 5°F	3.2.3	5204 6/
Abrasion resistance	3.2.3	5304 7/ 8/
Blocking	3.2.3	5872 8/

1/ Either side of the laminated cloth may be exposed to the light source. The exposure period shall be 100 hours.

2/ See 2.2.

3/ Either side of the laminated cloth may be exposed to the water in the hydrostatic tester.

4/ The side of the specimen which was exposed to the light source in the weathering unit shall be exposed to the water in the hydrostatic tester.

5/ The side of the specimen which was on the outside of the fold when rolled in the low temperature chamber shall be exposed to the water in the hydrostatic tester.

6/ The test specimens shall be subjected to a temperature of minus 65° ± 5°F for a minimum of 30 minutes and the test shall be performed in a still atmosphere at this temperature.

7/ Each side of the specimen shall be subjected to 50 cycles under a tension of 6 pounds, and under a load of 3 pounds pressure on the test abrader. The abradant shall be "Carborundum Fine Redi Cut J135R" emery cloth or equal.

8/ There shall be no failure of any test specimen.

* 4.5 Preparation for delivery examination. Examination of preparation for delivery shall be in accordance with the quality assurance provisions of PPP-P-1136.

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5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The laminated coated cloth, put-up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. The laminated coated cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with PPP-P-1136.

6. NOTES

6.1 Intended use. The laminated coated cloths covered by this document are intended for use in the manufacture of Bag, Waterproof, Signal Equipment MIL-B-10930.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) Type required (see 1.2).
(c) When a first article is required (see 3.1, 4.3 and 6.3).
(d) Whether color other than black is required (see 3.2.2.1).
(e) Width required (see 3.4).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of three yards of the finished cloth. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

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6.4 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-10351D(GL) Cloth, Laminated And Coated For Waterproof Containers

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